

Assignment 2 Report: Diagnostic Analysis of Healthcare Operations

May 2026

1. Introduction

This report diagnoses inefficiencies in hospital operations by linking patient characteristics with visit activity. The analysis uses two tables: Patients, with 400 records, and Visits, with 1,200 records from 2022 to 2025. The Tableau dashboard moves from operational overview to department, outcome, chronic-condition, city, and anomaly views so the analysis explains why patterns occur, not only what happened.

2. Data Modeling and Preparation

The dataset contains two source sheets. Patients includes Patient_ID, Age, Gender, City, and Chronic_Condition. Visits includes Patient_ID, Visit_Date, Department, Wait_Time_Minutes, Treatment_Cost, and Outcome. Patient_ID was used as the linking field and was treated consistently across both tables. In Tableau, the sheets were connected through a one-to-many relationship on Patient_ID, with Patients table including the primary key, and Visits including the foreign key. This preserves each table's grain: Patients is one row per patient and Visits is many rows per patient.

Of 400 patients, 383 have at least one recorded visit and 17 have no visit record. These 17 patients remain valid patient records, but they are excluded naturally from visit-based views because no visit activity exists to aggregate. Missing values were handled based on meaning. The 15 missing City values were retained through City Clean, which labels them as "Unknown." Missing city data does not invalidate the patient record. The 30 missing Wait_Time_Minutes values were not converted to zero because unknown wait and zero wait are operationally different. Instead, Valid Wait Flag excludes missing waits from wait-time views only, while the visits remain usable for cost, outcome, and volume analysis.

The workbook uses calculated fields for quality control and diagnosis: Has Visit Record, Missing Wait Count, Valid Wait Flag, City Clean, Admission Rate, Visits per Patient, Long Wait Flag, Wait Category, Cost per Minute, Missing Wait Rate, and Patient Count. These fields support coverage checks, missing-value tracking, utilization analysis, wait classification, and efficiency comparison. Three audit sheets - Patient Coverage, Missing Wait Times, and Missing Cities - validate the data preparation before dashboard interpretation.

3. Diagnostic Analysis

3.1 Wait Time Analysis

The overall valid wait time is about 93.7 minutes. Cardiology has the longest wait at 97.5 minutes, followed by Emergency at 93.6, Orthopedics at 92.3, and General at 91.4. The key diagnostic point is that Cardiology also has the lowest average treatment cost, about \$924, while the other departments range from about \$992 to \$1,066. This weakens the explanation that longer waits are driven by more expensive procedures. A more likely mechanism is a diagnostic-heavy workflow: patients wait for assessment, monitoring, or clinical decision-making, even when the final treatment is not high-cost.

3.2 Outcome Analysis

Outcome severity explains part of the delay but not all of it. Admitted patients wait 97.0 minutes on average, compared with 92.3 for Recovered patients and 92.1 for Follow-up patients. This is expected because admission decisions require fuller triage. However, the department-by-outcome view shows a stronger workflow issue. Cardiology Follow-ups average 104.4 minutes, while General Follow-ups average 82.1 minutes. This 22-minute gap inside one outcome group suggests that departmental process design, especially diagnostic review in Cardiology, is a major driver of wait variation.

3.3 Chronic Condition Impact

The chronic-condition pattern is counterintuitive. Among patients with at least one visit, chronic patients average 3.03 visits per patient, while non-chronic patients average 3.24. Their costs are also close: about \$1,008 for chronic patients versus \$995 for non-chronic patients. This does not match the usual expectation that chronic patients generate higher utilization. Possible explanations include scheduled chronic care outside this hospital system, mobility or access barriers, or a chronic flag that captures condition status but not severity. Resource planning should therefore avoid assuming that chronic patients are the main source of demand pressure in this dataset.

3.4 Regional Patterns

City patterns show both efficiency and data-quality concerns. Ottawa has the longest average wait at 96.3 minutes, followed by Calgary at 95.5, Toronto at 93.5, and Vancouver at 90.5. Ottawa also accounts for 12 of the 30 missing wait-time records, or 40% of all missing wait data. The combined pattern suggests pressure at intake: the same conditions that increase delay may also reduce documentation quality during busy periods.

3.5 Anomaly Detection

Two anomalies require attention. First, 209 visits have extended waits above 150 minutes. Cardiology and Emergency contain the largest counts, which fits their higher average waits and more complex triage needs. Second, 53 visits cost more than \$1,900. These are concentrated in General, with 18 cases, and Orthopedics, with 14 cases. General may receive mixed cases that should be routed earlier to specialists, while Orthopedics may create high costs through procedures, imaging, or materials. A third signal is age-related: patients under 30 wait about 99 minutes, while patients over 70 wait about 87 minutes, possibly because older patients arrive through clearer referral pathways.

4. Dashboard Design and Interactivity

The dashboard follows an overview-to-detail layout. It begins with KPI cards for total visits, average wait time, admission rate, total treatment cost, and missing wait rate. It then shows diagnostic panels for department wait time, department-by-outcome wait, chronic-condition comparison, city patterns, missing waits by city, and anomalies. The required Department, City, and Chronic Condition filters apply across the analysis. A Department highlight action allows users to click one department and cross-highlight its patterns across related views, making the dashboard useful for diagnosis rather than static reporting.

5. Key Insights

1. Cardiology has the highest wait time at 97.5 minutes but the lowest average cost at about \$924. This suggests diagnostic delay rather than procedure cost. Wait time and cost should be monitored separately because a low-cost department can still create patient-flow delays.
2. Admitted patients wait longer than other outcomes: 97.0 minutes versus 92.3 for Recovered and 92.1 for Follow-up. This likely reflects more complete triage before admission. Outcome severity matters, but it does not explain all department-level variation.
3. Follow-up visits show the strongest workflow gap. Cardiology Follow-ups average 104.4 minutes, while General Follow-ups average 82.1. Department-specific follow-up protocols are therefore a better improvement target than a broad wait-time campaign.
4. Chronic patients do not visit more often in this hospital data. They average 3.03 visits per patient compared with 3.24 for non-chronic patients, and costs are similar at \$1,008 versus \$995. Capacity planning based only on assumptions about chronic utilization may misallocate resources.

5. Ottawa has the longest average wait at 96.3 minutes and produces 40% of missing wait-time records. This points to a combined service and documentation issue. Reliable measurement should come before major staffing changes.

5. Recommendations

1. Schedule Cardiology Follow-up visits in the first half of the day, when diagnostic and lab capacity is more likely to be available. Intake staff should pre-pull EKG, lab, and previous-visit information before assessment. This targets the 104.4-minute Cardiology Follow-up average and should move it closer to the overall Follow-up average of about 92 minutes.
2. Audit Ottawa intake and make wait-time entry mandatory before a visit can be closed. This addresses both the highest city wait and the 12 missing Ottawa wait records. Better data capture should happen before staffing changes because improvement cannot be verified with incomplete wait data.

6. Limitations

The dataset is useful for diagnosis, but several group differences are small. Findings should be interpreted as operational signals, not proven causes. Testing the explanations would require additional data such as staffing levels, arrival time by hour, triage category, procedure logs, and room capacity.

7. AI Tool Usage Disclosure

ChatGPT was used as an analytical and writing-support tool. Specifically, it was used to review the assignment requirements and rubric, verify summary statistics from the Patients and Visits data, organize the report into the required sections, and improve the wording of findings using a Finding, Why, Implication structure. The Tableau workbook, dashboard layout, calculated fields, filters, and dashboard actions were created by the group. The final report should be reviewed by the group before submission to confirm that all wording matches the submitted dashboard and the group's interpretation.